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Exploring coastal ecosystem adaptation with experimental arrays

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12:30 – 1:50 PM
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Ecotonal coastal wetlands are experiencing rapid vegetation expansion and contraction, increased hydrological energy and submergence, and can serve as a bellwether for other rapidly changing ecosystems. Predicting the responses of marshes and mangroves, and their soil processes, to these changes can be improved by pushing ecosystems using experimental arrays. Through the WETFEET project, we have been performing large scale manipulations of air temperatures, inundation and nutrient availability at multiple wetland sites across Northeast Florida. We have shown that mangroves respond subtly to experimentally imposed chronic warming and grew slightly more with warming near their northern limit. However, the strongest response of mangroves to warming is manifested on the landscape scale, as declining frequency of episodic freeze events has allowed their expansion to gain inertia, at the expense of marshes. Adding nitrogen to mangroves and marsh plants across multiple hydrological settings helped us explain contradictory results in previous studies of wetland response to nitrogen addition. Specifically, nitrogen addition stimulated root growth in the wetland interior, where nitrogen is more strongly limiting, but reduced root growth along the water's edge. Adding nitrogen also diminished wetland plant and soil nitrogen sinks. As we conduct these experiments, our very plots have been eroding and submerging into the sea, illustrating that land loss, rather than shifting temperatures or excess nutrients, is likely the greatest threat to viability of these, and other, coastal wetlands. Our work continues to focus on examining coastal wetland erosion and the nutrient sink implications of this land loss.